

Modelling & Simulation: Final Assignment

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1 Introduction

The objective of this assignment was to design a robotic manipulator capable of accurately tracking Cartesian position references while maintaining compliant behaviour during external interactions. One of the early choices was the type of robot and application. In our case, the robot was designed as a lightweight desktop manipulator made of aluminium, which contained series elastic actuators (SEAs) in each joint to improve safety and compliance.

The main challenge was the complex controller architecture: Cartesian impedance control for trajectory, torque control based on SEAs deflection, current control for actuator dynamics and gravity compensation to reduce the effect of the arm weight on the performance. To tackle this, some submodels were developed to independently tune each controller and some simulations were also carried out to assess performance under external disturbances.

2 Model

2.1 Theoretical background

2.2 Forward kinematics

To translate the angles the modeled arm would return into cartesian coordinates, the forward kinematics problem needs to be solved. The normal forward kinematics given the directions of the angles as defined in the 3D model for the two DoF arm is

$$\begin{aligned}r &= L_1 \cos(q_2) + L_2 \cos(q_2 + q_3) \\z &= -L_1 \sin(q_2) - L_2 \sin(q_2 + q_3)\end{aligned}$$

This 2 DoF arm gives the position in 3D after applying cylindrical coordinates.

$$\begin{aligned}x &= r \cos(q_1) \\y &= r \sin(q_1)\end{aligned}$$

2.2.1 Cartesian Impedance Control

To do impedance control in cartesian coordinates, we require an error expressed in cartesian coordinates to calculate a desired force with a PD-controller. This force can then be mapped to joint torques via the Jacobian.

$$\tau = J^T(k_p \odot e + k_d \odot \dot{e}) \quad (1)$$

where $k_p, k_d, e \in \mathbb{R}^3$ and $\odot$ denotes element-wise multiplication. The error can be calculated with forward kinematics

$$e = x_{SP} - f(q)$$

where $x_{SP}, q \in \mathbb{R}^3$ is the desired end effector location and the joint angles, respectively.

2.2.2 Torque Control

It is used a Series Elastic Actuator (SEA) torque control loop, which measures the torsional spring deflection of the SEA (located between the motor and the load) and converts it into torque using Hooke's law:

$$\tau_m = K_s(q_l - q_m) \quad (2)$$

where τ_m is the transmitted torque, K_s is the spring stiffness and $(q_l - q_m)$ is the angle difference between the load and the motor, also known as spring deflection. This torque is then compared to the reference one to compute the tracking error:

$$e = \tau_{set} - \tau_m$$

that serves as the input of a PD controller which adjusts the current that will be injected in the following current controller.

2.2.3 Current Control

To control the current, a PI controller is used. The integral term assures zero steady-state error and due to the fast dynamics the proportional term can handle quick reactions. A derivative term is not used because motors in reality use PWM signals to apply voltages and the current sensors are encoded and noisy. This would make any estimate of the derivative of current practically unusable. For feedforward, we additively inject the desired (electrical) dynamics.

$$v_{ff} = (R_h + R_m)i_d + L \frac{di_d}{dt} \quad (3)$$

where R_h is the H-bridge resistance, R_m is the motor resistance, and L is the inductance. The friction could have been pulled through the gyrator to obtain its equivalent impedance of the other side, but the resulting term was unnaturally high (division by small number).

Since the current loop is the fastest, it will get a new set point every couple (around 5 to 10) samples. Therefore, the derivative of the desired current will be a delta function, which could lead to overshoot in the inductance compensating term.

This feed-forward term does not take any static behavior of the plant into account because that is already covered by the gravity compensation term. In order to take the p-type storage element (motor and gearbox inertia) between the gyrator and transformer into account (because it acts as a q-type storage element on the other side of the gyrator), we would need to be able to measure or approximate its state. While this is not impossible, it would be difficult to do accurately with the noisy current measurement (integral of current may drift or be inaccurate). So, this term is neglected from the feed-forward control law.

2.2.4 Gravity Compensation

In order to compensate gravity, we must first find the potential energy terms of the robotic arm that correspond with the potential field due to gravity. Using forward kinematics to find the height of each bodies center of mass gives us

$$h_1 = k$$

$$h_2 = k - L_{c,2} \sin(q_2)$$

$$h_3 = k - L_2 \sin(q_2) - L_{c,3} \sin(q_2 + q_3)$$

where h_i , L_i , and $L_{c,i}$ are the height, length and distance from joint i to the center of mass of body i for body i , respectively. The potential energy due to gravity can be written as

$$V_g = \sum_i m_i g h_i$$

The gradient of this potential energy will be additively fed into the system

$$\tau_{ctrl} = \tau + \frac{dV_g}{dq}$$

where τ is the result of the impedance control law as described in Eq. 1. The gradient is

$$\frac{dV_g}{dq} = -g \begin{bmatrix} 0 \\ (m_2 L_{c,2} + m_3 L_2) \cos(q_2) + m_3 L_{c,3} \cos(q_2 + q_3) \\ m_3 L_{c,3} \cos(q_2 + q_3) \end{bmatrix}$$

The reason gravity compensation is added after impedance control is that the torque contribution due to the gradient needs to be applied at the joint (i.e. after SEA). Thus, the torque controller should also enforce gravity compensation. Since the SEA acts as a low-pass filter, a feedforward gravity compensation term would not be bothered by the SEA (unless the set point changes frequently). We use feedback gravity compensation as it is more accurate in removing the effect of gravity. Otherwise, the rendered stiffness from the impedance controller would not function as well.

2.3 Model overview

This model implements a cascaded control loop for a robotic arm by stacking three discrete controllers: impedance, torque and current. Each of them operates at a different sampling frequency, being the outer impedance loop the slowest one

and the inner current controller the fastest one. In addition, gravity compensation is included. All the control part is run in discrete, while the physical components operate in continuous. This model allows the selection of desired position in cartesian coordinates and the addition of some disturbance at the end effector.

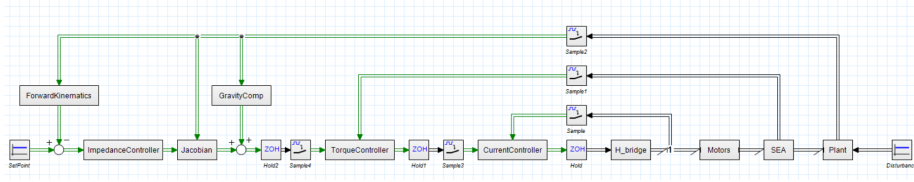


Figure 1: Overview of "G09_Final Assignment v8.emx"

The plant block contains the 3D-3DOF robot arm with its rotational joints, which was modeled using the 3D Mechanics Editor. The decided orientation is yaw-pitch-pitch.

2.4 Modelling choices

2.4.1 Parameters of the robot links and selection of motor and gearbox

It was decided to model a small desktop robot arm intended to pick-and-place light objects or hold a small camera or sensor. Based on robot models used for these applications, similar values were chosen for the link lengths. The main dimensions and parameters used in the model are summarized in Table 1.

The arm links were modeled as hollow cylinders to reduce weight and accommodate wiring. Aluminium was selected as the structural material due to its strength, low density (approximately 2700 kg/m^3) and widespread use in robotic systems. Using the dimensions listed in Table 1, the mass of each link was estimated from the volume of the hollow links and material density.

Parameter	Value
Base link radius	0.05 m
Base link height	0.25 m
Forearm link outer radius	0.025 m
Forearm link length	0.25 m
Lower arm link outer radius	0.025 m
Lower arm link length	0.30 m
Wall thickness of all links	0.0025 m
Aluminium density	2700 kg/cm^3

Table 1: Model parameters for the robotic arm

Then, a suitable motor and gearbox combination had to be selected. The two main candidates were the Maxon and Nanotec combinations used in Assignment 2. The Nanotec motor and gearbox assembly was found to be extremely heavy for a desktop robotic arm, with a combined mass of 1.6 kg. In contrast, the

Maxon combination weighs only 0.420 kg, making it a more suitable option for this purpose.

2.4.2 Stiffness of the SEA springs

The SEA was modeled as a simple torsional spring placed between the actuator output and the robot joint. A C element was placed through a 0-junction. Only the elastic behavior of the SEA was considered, so damping effects were neglected and the torque transmitted through the SEA is proportional to the spring deflection.

The stiffness of the SEA springs was selected based on the desired compliance of the robotic arm during human interaction. Rather than maximizing joint rigidity, the objective was to allow a considerable deflection when external forces are applied in order to reduce the risk of injury.

To determine suitable stiffness values, each joint was evaluated independently using the following experiment. Without motors or controllers, the SEAs and plant were isolated and a force of 20 N was applied at the end effector while the remaining two joints were held in place. Gravity was also turned off. The spring stiffness of the joint under consideration was then adjusted until the maximum angular deflection was approximately $\pi/4$ rad, which was a value that seemed reasonable. The obtained stiffness values were 28, 28 and 13 Nm/rad respectively.

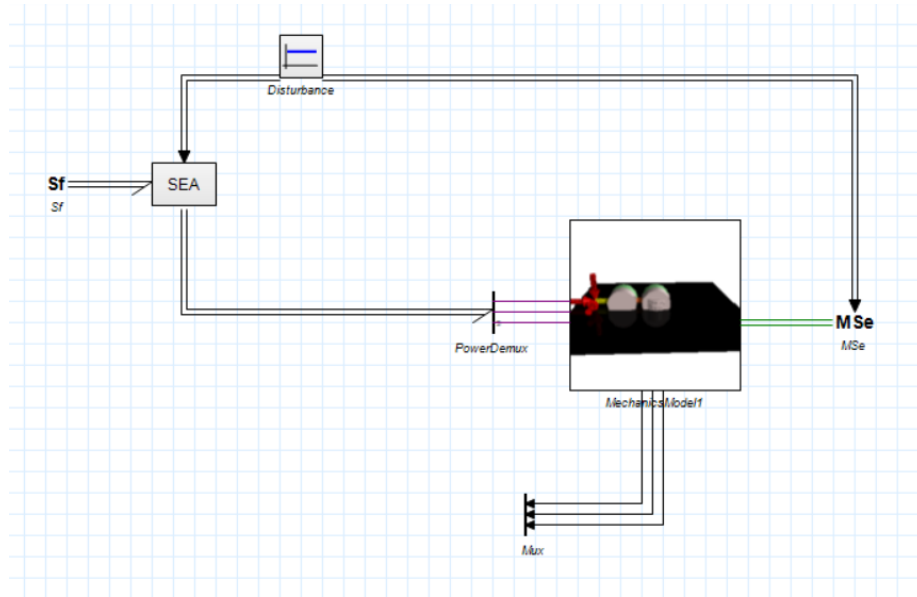


Figure 2: SEA stiffness model in "G09_joint_stiffness_chooser.emx"

2.4.3 Viscous and Coulomb friction

In order to implement a realistic friction model, both viscous and Coulomb friction must be taken into account. To do so, a similar procedure was carried

out individually in the motor and the gearbox. Firstly, the resistance parameter R needed to be found.

For the motor, R was tuned such that the model matched the maximum efficiency point at nominal voltage and torque. This was done by imposing the operating torque on one extreme and using the known efficiency at that condition to solve for R . The resulting value ensures that the electrical losses represented by the model are consistent with the peak-efficiency operating point.

For the gearbox, a similar approach was used. The resistance parameter R was calibrated to match the maximum gearbox efficiency at both maximum torque and maximum rotational speed. In this case, both were enforced at their respective extreme operating points.

After determining the equivalent resistance values for both components, these losses were decomposed into viscous and Coulomb friction components. It was assumed that approximately 90% of the losses could be attributed to Coulomb friction, while the remaining 10% were consequence of viscous friction effects.

The final friction model was implemented through the following equation:

$$\tau_f = d_c \tanh(5000 \omega) + d_v \omega \quad (4)$$

2.4.4 Current Controller

Since the current controller was the one running at highest frequency being the inner loop, it was decided to isolate it with the motor to perform an initial tuning by locking the output (forced zero velocity) and inputting step signals as current. The current controller contains a feedforward term which was able to effectively reduce the error in less than five samples.

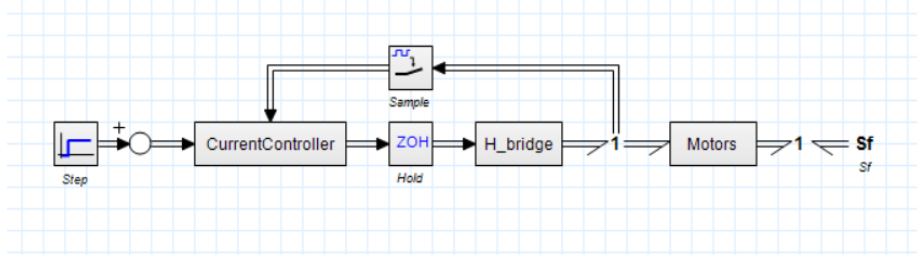


Figure 3: Current controller tuning model in "G09_current_controller2.emx"

3 Simulation

3.1 Gravity compensation

If gravity is compensated perfectly and the compensation is the only force given to the system, then the robotic arm will not move in any configuration. To this end, the gravity compensation term will be applied directly to the joints, and the initial configuration will be varied.

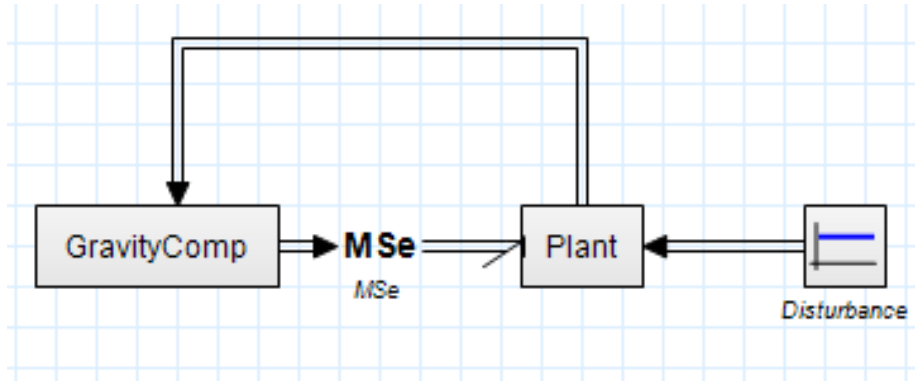


Figure 4: Gravity Compensation Model in "G09_Gravity Comp Check.emx"

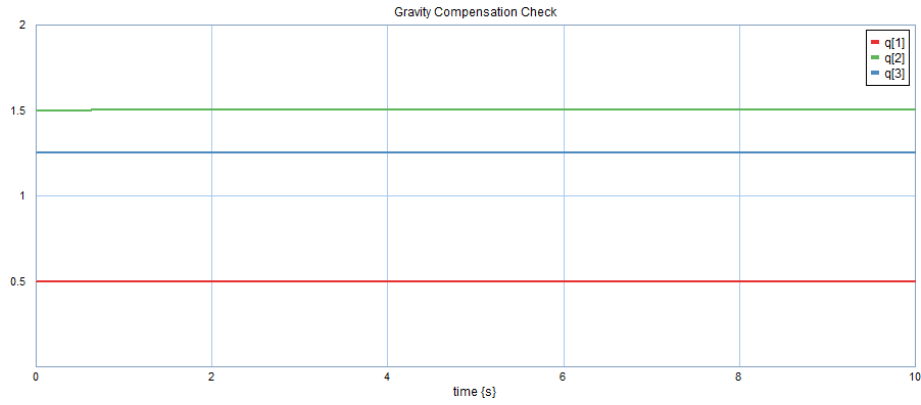


Figure 5: Gravity Compensation Validation in "G09_Gravity Comp Check.emx"

Regardless of which initial configuration is chosen, the angles remain constant.

3.2 Current Controller

All the simulations shown below were run using the "G09_current_controller2.emx" tuning model. The step response of the current controller in continuous time is given in Fig. 6. The crosses in the set point current indicate simulation time steps. Due to the extraordinarily fast dynamics, the integration method's error was reduced to $1e-9$ (relative and absolute). Observing the graph, it is clear that the controller reaches the set point within $20 \mu s$. If we assume a sampling frequency of 10 kHz, we get a sampling time of $1000 \mu s$, which means the continuous controller reaches the set point in less than one sample of the discrete controller. This performance is deemed sufficient and the controller is therefore discretized.

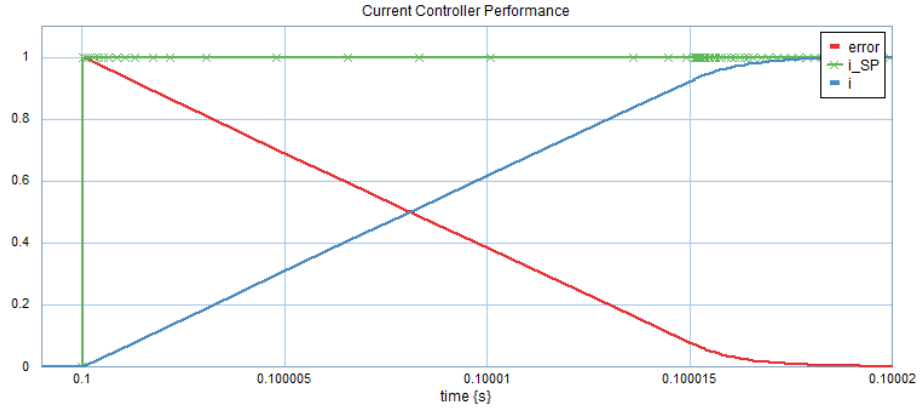


Figure 6: Performance of the current controller in continuous time

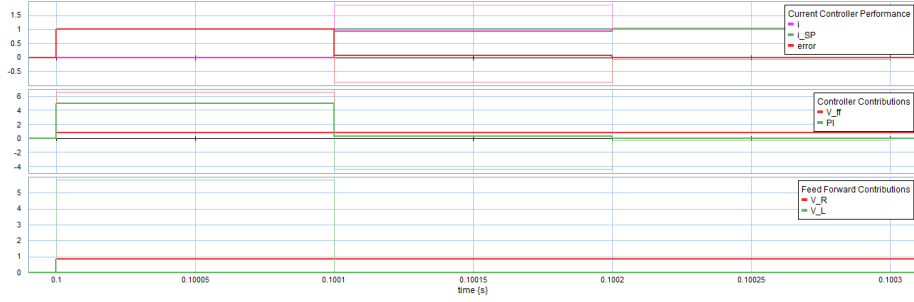


Figure 7: Comparison of the controller's performance with (faded lines) and without (solid lines) inductance compensating feed forward term

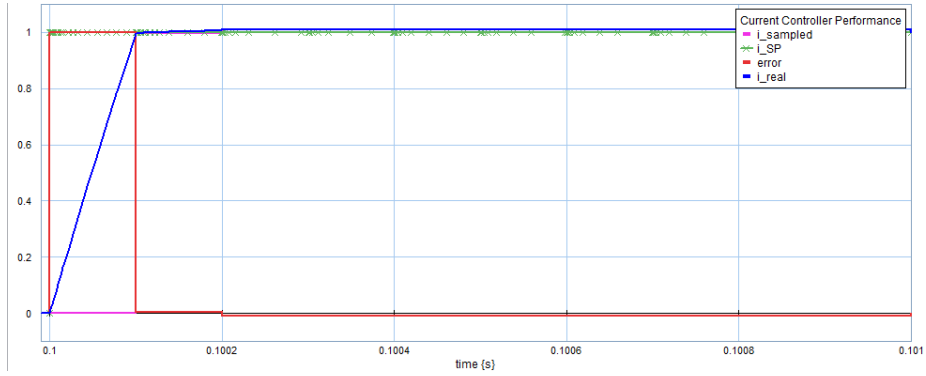


Figure 8: The performance of the tuned current controller

Fig. 7 tells us that the inclusion of the inductive term in the feed forward is not beneficial and was excluded from the model that generated Fig. 8.

3.3 Torque Controller

For an initial tuning of the torque controller, the plant was removed due to its extreme non-linearity and the current controller was assumed to be ideal.

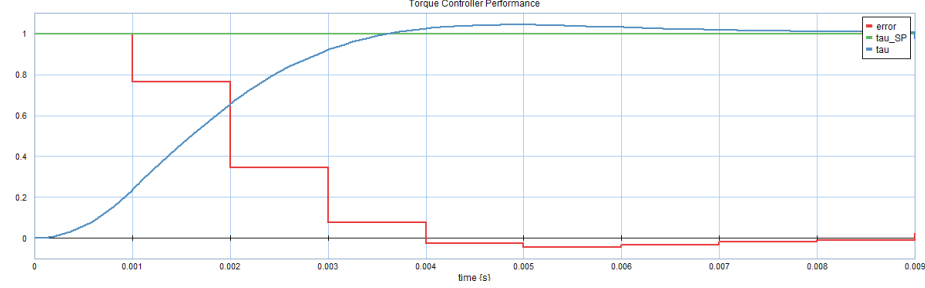


Figure 9: Initially tuned torque controller's response without plant (with a stiffness of 10 Nm/rad) in "G09 _"

After initial tuning, a pretty decent torque response was obtained. However, when adding it to the whole model, it completely stopped working. A lot of noise and oscillations were present. While trying to find out the reason, it was observed that the model would break if more than just a step response was given. Despite this, the torque controller still had a very quick response. Eventually, some damping was added to the joints in order to make the system more consistent when there were quickly changing values.

3.4 Impedance and Current Control

When the impedance control loop runs at 250 Hz and the current control loop at 10 kHz (torque control would have been at 2 kHz), the system is able to track a point nicely as it can be seen in the figure below. This simplification excludes the torque controller and SEA, and is meant to test the impedance and current control combination as a unit.

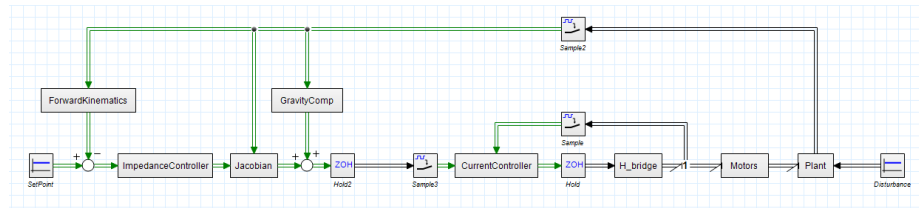


Figure 10: Model "G09_ImpedanceCtrl+CurrentCtrl model.emx" used for the generation of figures in sections 3.4 and 3.5.

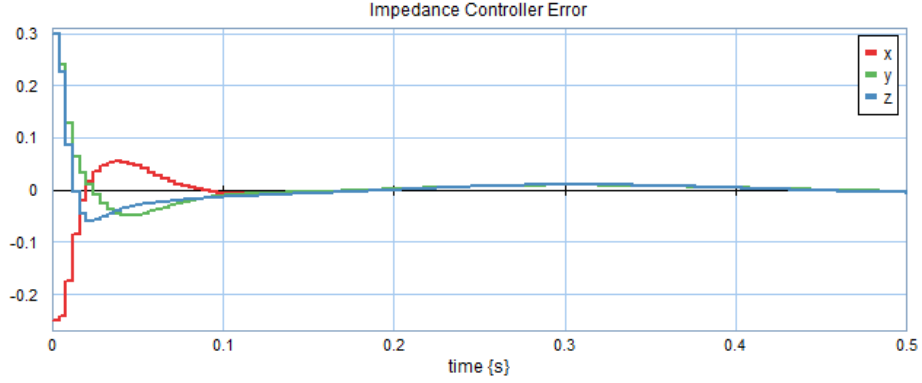


Figure 11: Impedance control with current control moving the end effector to the point $[0.3, 0.3, 0.3]$ meters without disturbance.

The steady-state error in this case is about $-711 \mu\text{m}$ in x and y direction, and 1.99 mm in z direction. After reaching steady-state, a disturbance of 5 N is given in each direction one at a time. The disturbance is introduced at 10 seconds and is expressed in the body fixed frame of the end effector.

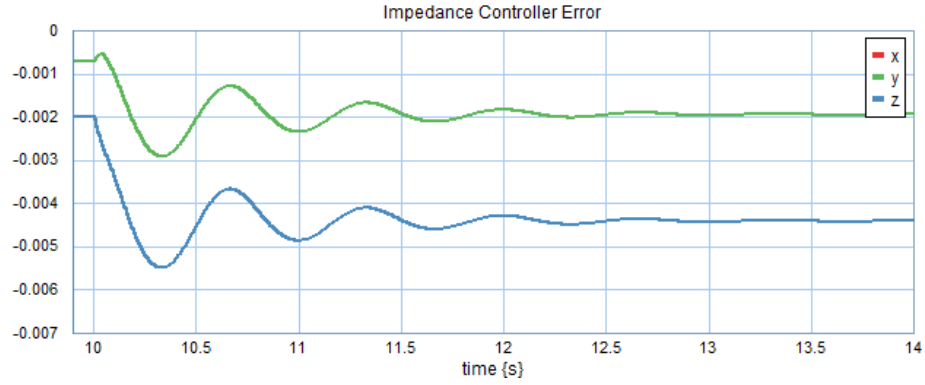


Figure 12: Impedance control with current control moving the end effector to the point $[0.3, 0.3, 0.3]$ meters with a disturbance of 5 N in x-direction (note that x and y overlap).

The steady-state error is about -1.94 mm in x and y direction and -4.41 mm in z direction.

When giving the same disturbance in the y-direction, the z-coordinate remains unbothered, while the other two coordinates move in opposite direction from their undisturbed steady state error. This is because the disturbance in y-direction pushes in such a way that only the local y-rotation joints actually works (no torque on z-rotation joint).

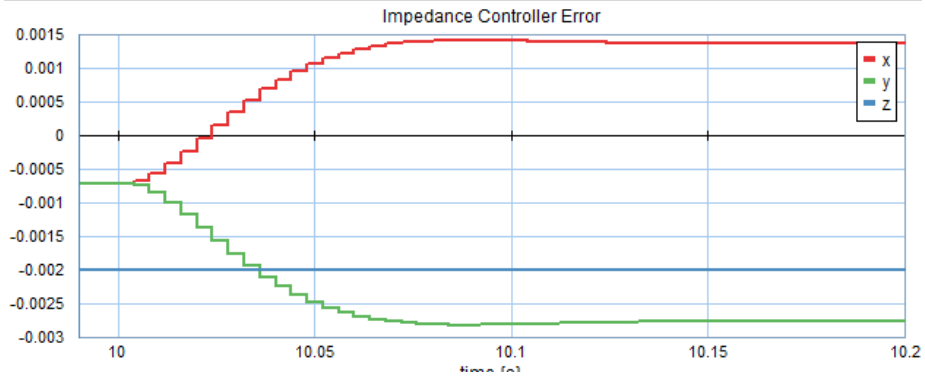


Figure 13: Impedance control with current control moving the end effector to the point $[0.3, 0.3, 0.3]$ meters with a disturbance of 5 N in y-direction.

The disturbance in z-direction gives a response that causes x and y to overlap.

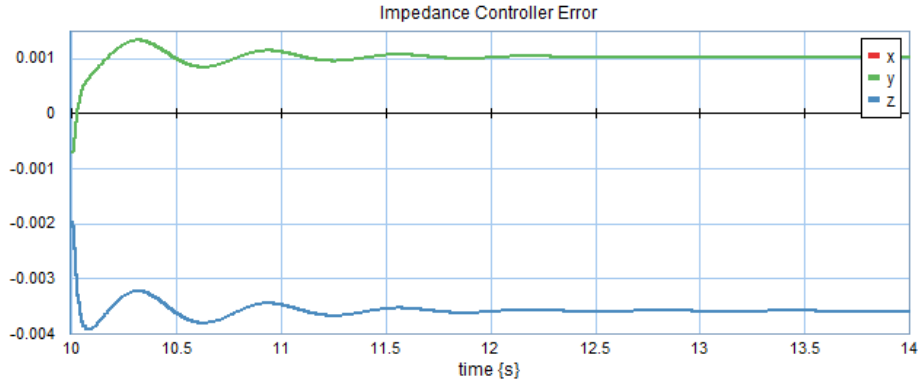


Figure 14: Impedance control with current control moving the end effector to the point $[0.3, 0.3, 0.3]$ meters with a disturbance of 5 N in z-direction.

The x and y direction's behavior overlaps in two of the three cases due the disturbance being in a local frame, which caused symmetry. Fig. 12 and Fig. 14 show that the system is underdamped in some directions. This is because the damping ratio depends on the observed mass and inertia, which is different in different directions. The reason several coordinates change is also due to the fact that the robotic arm is kinematically constraint. A force in some direction will translate to a torque on possibly all joints via the Jacobian.

3.5 Final model

It was achieved a working model with a functional controller only after removing the imposed constraints, which were causing a lot of problems. This unconstrained model is able to track the reference trajectory (an 8 path) in an accurate manner and manages disturbances of 10 N every 10 seconds in all 3 directions in a decent manner, as shown below:

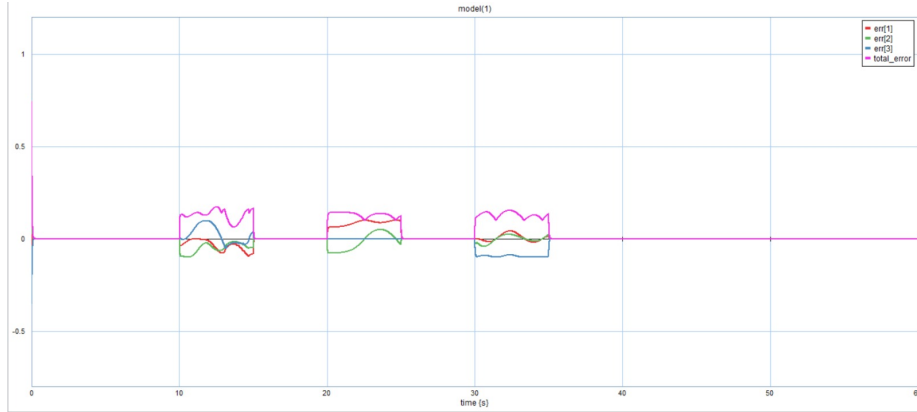


Figure 15: End-effector error in Cartesian from "G09_impedance_controller_tuning.emx"

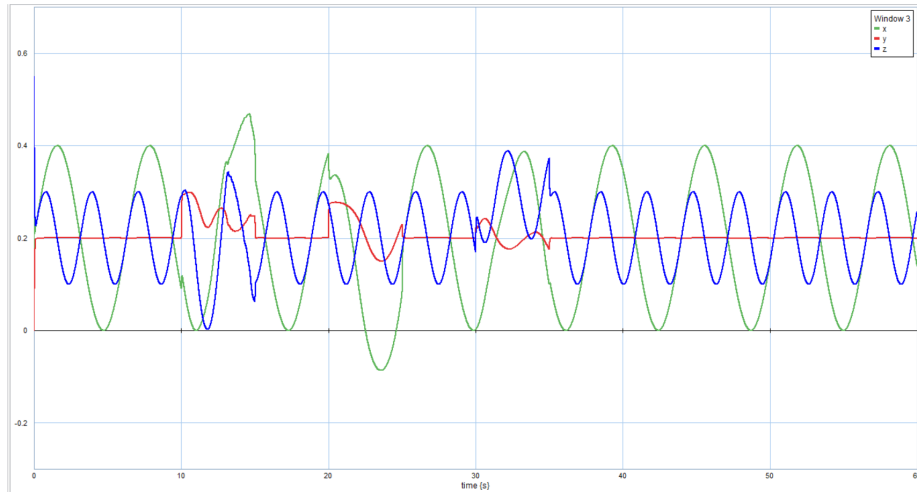


Figure 16: End-effector position with disturbances in Cartesian from "G09_impedance_controller_tuning.emx"

4 Discussion & Conclusion

The initial tuning of the controllers was performed by observing how changing the gain values affected the error. While this approach works, it does not guarantee an ideal response as more formal approaches would do. During the torque controller (PD) tuning, an initially poorer response was accepted for the sake of lower steady-state error. Simultaneously, the gains were kept as low as possible without affecting much the performance of the controllers. While this was done for stability, interconnecting the loops did result in some issues. Most notably, the instability after insertion of the torque controller. Some simulation results were recorded without the torque controller and SEAs (basically infinitely stiff) for this reason. The results yielded by this approach were deemed satisfactory.

Furthermore, due to late changes to e.g., the stiffness of the SEAs and the 3D mechanics file, controllers had to be retuned several times and gravity compensation had to be readjusted (20-sim's global variables were not working among other software bugs). This cost valuable time that would have been needed to fix the torque controller induced instability.

We also tried to use critical damping estimation to calculate a start for the damping. This was done by measuring the acceleration of the end effector when pushing on one limb. An steady acceleration was obtained; however, the calculated damping, using $F = m \cdot A$ and $damping = 2 \cdot \sqrt{2 \cdot m}$, was extremely low.

To conclude, we have learned a lot about compromises that have to be made in order to tune controllers, which still may end up in instabilities in the whole model. A solution we came across (but did not implement because of time constraints), was to calculate the effective inertias of the plant at several points and use those values to tune several controllers among which the system could swap based on the current configuration.